



MS/MA IN ECONOMICS

DYNAMIC STOCHASTIC MODELS

Lecture 08

Solving the Matrix Quadratic Equation

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1 The problem

We are asked to solve the matrix quadratic equation

$$AP^2 + BP + C = 0, \quad (1)$$

where A , B and C are square matrices of the same size that we are free to choose, and P is the unknown matrix of the same dimension.

2 Choosing the matrices

We work with $n = 2$. We take

$$A = \begin{bmatrix} 3 & 0.5 \\ 7 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}, \quad C = \begin{bmatrix} -1.25 & -0.02 \\ -2.75 & -0.24 \end{bmatrix}. \quad (2)$$

The only requirement the method imposes on the data is that A be invertible: it enters the pair (D, E) in a position where a singular A would produce infinite generalized eigenvalues and break the construction of P . We check this directly,

$$\det A = (3)(1) - (0.5)(7) = 3 - 3.5 = -0.5 \neq 0, \quad (3)$$

so A is invertible. No condition is placed on B or C . From here on we treat P as unknown and recover it from A , B , C alone.

3 The method

The idea is to turn (1) into a generalized eigenvalue problem. Following the text, from the $n \times n$ matrices A , B , C we build the $2n \times 2n$ matrices ¹

$$D = \begin{bmatrix} -B & -C \\ I & 0 \end{bmatrix}, \quad E = \begin{bmatrix} A & 0 \\ 0 & I \end{bmatrix}, \quad (4)$$

where I and 0 are the $n \times n$ identity and zero blocks. We then solve the generalized eigenvalue problem for the pair (D, E) : we look for scalars λ_k and vectors x_k such that

$$D x_k = E x_k \lambda_k. \quad (5)$$

¹The text builds D from B and C for the equation $AP^2 - BP - C = 0$. Since we solve $AP^2 + BP + C = 0$, we substitute $B \rightarrow -B$ and $C \rightarrow -C$, which is why D carries the blocks $-B$ and $-C$ below.

This yields $2n$ eigenvalues. The stable ones, those with $|\lambda| < 1$, correspond to dynamics that die out and return to the steady state; the unstable ones ($|\lambda| > 1$) explode. Only the stable part gives an economically sensible solution, so we keep the n eigenvalues with $|\lambda| < 1$ (the n smallest in modulus if there are more than n). This is the matrix counterpart of choosing the root with $|P| < 1$ in the scalar case.

Ordering the stable eigenvalues first and partitioning the matrix of generalized eigenvectors as

$$X = \begin{bmatrix} X_{11} & X_{21} \\ X_{12} & X_{22} \end{bmatrix}, \quad (6)$$

the solution is built from the lower block of the stable eigenvectors, X_{12} , and the diagonal matrix of stable eigenvalues, Δ_1 , through

$$P = X_{12} \Delta_1 X_{12}^{-1}. \quad (7)$$

Writing P this way gives $P^2 = X_{12} \Delta_1^2 X_{12}^{-1}$, which is what makes it fit the quadratic.

4 Building D and E

With our matrices, the blocks are

$$-B = \begin{bmatrix} -1 & 0 \\ -2 & -1 \end{bmatrix}, \quad -C = \begin{bmatrix} 1.25 & 0.02 \\ 2.75 & 0.24 \end{bmatrix}, \quad (8)$$

so that

$$D = \begin{bmatrix} -1 & 0 & 1.25 & 0.02 \\ -2 & -1 & 2.75 & 0.24 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}, \quad E = \begin{bmatrix} 3 & 0.5 & 0 & 0 \\ 7 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}. \quad (9)$$

5 The generalized eigenvalues

The generalized eigenvalues solve $\det(D - \lambda E) = 0$. Forming λE and subtracting,

$$D - \lambda E = \begin{bmatrix} -1 - 3\lambda & -0.5\lambda & 1.25 & 0.02 \\ -2 - 7\lambda & -1 - \lambda & 2.75 & 0.24 \\ 1 & 0 & -\lambda & 0 \\ 0 & 1 & 0 & -\lambda \end{bmatrix}. \quad (10)$$

We expand the determinant along the third row $(1, 0, -\lambda, 0)$, which has two zeros. With the sign

pattern $(+, -, +, -)$ only the first and third entries contribute:

$$\det(D - \lambda E) = 1 \cdot \det(M_{31}) + (-\lambda) \cdot \det(M_{33}), \quad (11)$$

where M_{31} and M_{33} are obtained by deleting row 3 and the corresponding column,

$$M_{31} = \begin{bmatrix} -0.5\lambda & 1.25 & 0.02 \\ -1 - \lambda & 2.75 & 0.24 \\ 1 & 0 & -\lambda \end{bmatrix}, \quad M_{33} = \begin{bmatrix} -1 - 3\lambda & -0.5\lambda & 0.02 \\ -2 - 7\lambda & -1 - \lambda & 0.24 \\ 0 & 1 & -\lambda \end{bmatrix}. \quad (12)$$

Computing $\det(M_{31})$. Expanding along its third row $(1, 0, -\lambda)$,

$$\det(M_{31}) = 1 \cdot \det \begin{bmatrix} 1.25 & 0.02 \\ 2.75 & 0.24 \end{bmatrix} + (-\lambda) \cdot \det \begin{bmatrix} -0.5\lambda & 1.25 \\ -1 - \lambda & 2.75 \end{bmatrix} \quad (13)$$

$$= [(1.25)(0.24) - (0.02)(2.75)] - \lambda [(-0.5\lambda)(2.75) - (1.25)(-1 - \lambda)] \quad (14)$$

$$= [0.30 - 0.055] - \lambda [-1.375\lambda + 1.25 + 1.25\lambda] \quad (15)$$

$$= 0.245 - \lambda [1.25 - 0.125\lambda] \quad (16)$$

$$= 0.125\lambda^2 - 1.25\lambda + 0.245 \quad (17)$$

Computing $\det(M_{33})$. Expanding along its third row $(0, 1, -\lambda)$,

$$\det(M_{33}) = -1 \cdot \det \begin{bmatrix} -1 - 3\lambda & 0.02 \\ -2 - 7\lambda & 0.24 \end{bmatrix} + (-\lambda) \cdot \det \begin{bmatrix} -1 - 3\lambda & -0.5\lambda \\ -2 - 7\lambda & -1 - \lambda \end{bmatrix}. \quad (18)$$

The first 2×2 determinant is

$$(-1 - 3\lambda)(0.24) - (0.02)(-2 - 7\lambda) = -0.24 - 0.72\lambda + 0.04 + 0.14\lambda = -0.20 - 0.58\lambda, \quad (19)$$

and the second is

$$(-1 - 3\lambda)(-1 - \lambda) - (-0.5\lambda)(-2 - 7\lambda) = (1 + 4\lambda + 3\lambda^2) - (\lambda + 3.5\lambda^2) \quad (20)$$

$$= 1 + 3\lambda - 0.5\lambda^2. \quad (21)$$

Therefore

$$\det(M_{33}) = -(-0.20 - 0.58\lambda) - \lambda(1 + 3\lambda - 0.5\lambda^2) \quad (22)$$

$$= 0.20 + 0.58\lambda - \lambda - 3\lambda^2 + 0.5\lambda^3 \quad (23)$$

$$= 0.5\lambda^3 - 3\lambda^2 - 0.42\lambda + 0.20 \quad (24)$$

Assembling the determinant. Combining the two pieces,

$$\det(D - \lambda E) = (0.125\lambda^2 - 1.25\lambda + 0.245) - \lambda(0.5\lambda^3 - 3\lambda^2 - 0.42\lambda + 0.20) \quad (25)$$

$$= -0.5\lambda^4 + 3\lambda^3 + 0.545\lambda^2 - 1.45\lambda + 0.245. \quad (26)$$

Setting this to zero and multiplying by -2 (which does not change the roots) gives the characteristic polynomial in clean form,

$$\lambda^4 - 6\lambda^3 - 1.09\lambda^2 + 2.9\lambda - 0.49 = 0. \quad (27)$$

6 Finding the roots

Equation (27) is a quartic. Rather than reach for a closed-form quartic formula, we locate the roots by evaluating the polynomial $p(\lambda) = \lambda^4 - 6\lambda^3 - 1.09\lambda^2 + 2.9\lambda - 0.49$ and tracking sign changes, since a change of sign between two points brackets a root. To know where to search, we first bound the roots. For a monic polynomial the Cauchy bound states that every root satisfies $|\lambda| < 1 + \max_i |a_i|$, where the a_i are the remaining coefficients; here $\max_i |a_i| = 6$, so all four roots lie in $|\lambda| < 7$. It is therefore enough to sweep the interval $(-7, 7)$: any root must show up as a sign change within it.

λ	-1	-0.5	0	0.2	0.5	6	7
$p(\lambda)$	2.52	-1.40	-0.49	0.00	0.00	-22.33	309.40

Table 1: Values of the characteristic polynomial. Sign changes bracket roots in $(-1, -0.5)$ and $(6, 7)$; the polynomial vanishes at $\lambda = 0.2$ and $\lambda = 0.5$.

The evaluation flags $\lambda = 0.2$ and $\lambda = 0.5$ as candidates, since p is essentially zero there. We verify both exactly. At $\lambda = 0.2$,

$$(0.2)^4 - 6(0.2)^3 - 1.09(0.2)^2 + 2.9(0.2) - 0.49 = 0.0016 - 0.048 - 0.0436 + 0.58 - 0.49 = 0, \quad (28)$$

and at $\lambda = 0.5$,

$$(0.5)^4 - 6(0.5)^3 - 1.09(0.5)^2 + 2.9(0.5) - 0.49 = 0.0625 - 0.75 - 0.2725 + 1.45 - 0.49 = 0. \quad (29)$$

Both are roots, and both have modulus below one, so they are the two *stable* eigenvalues we need. The remaining two roots, bracketed in $(-1, -0.5)$ and $(6, 7)$, are unstable (numerically $\lambda \approx -0.803$ and $\lambda \approx 6.103$) and play no role in constructing P .

7 The eigenvectors

For each stable eigenvalue we solve $(D - \lambda E)x = 0$ for the associated generalized eigenvector $x = (x_1, x_2, x_3, x_4)^\top$. The last two rows of $D - \lambda E$ read

$$x_1 = \lambda x_3, \quad x_2 = \lambda x_4, \quad (30)$$

so the upper half of x is λ times the lower half; this holds by construction of D and is what makes the method work.

Eigenvalue $\lambda = 0.2$. Here $x_1 = 0.2 x_3$ and $x_2 = 0.2 x_4$. Using the first row,

$$-1.6 x_1 - 0.1 x_2 + 1.25 x_3 + 0.02 x_4 = 0, \quad (31)$$

and substituting,

$$-1.6(0.2 x_3) - 0.1(0.2 x_4) + 1.25 x_3 + 0.02 x_4 = 0.93 x_3 = 0 \Rightarrow x_3 = 0, x_1 = 0. \quad (32)$$

With x_4 free, set $x_4 = 1$, so $x_2 = 0.2$. The eigenvector is

$$v_{0.2} = \begin{bmatrix} 0 & 0.2 & 0 & 1 \end{bmatrix}^\top. \quad (33)$$

Eigenvalue $\lambda = 0.5$. Here $x_1 = 0.5 x_3$ and $x_2 = 0.5 x_4$. Using the first row,

$$-2.5(0.5 x_3) - 0.25(0.5 x_4) + 1.25 x_3 + 0.02 x_4 = -0.105 x_4 = 0 \Rightarrow x_4 = 0, x_2 = 0. \quad (34)$$

With x_3 free, set $x_3 = 1$, so $x_1 = 0.5$. The eigenvector is

$$v_{0.5} = \begin{bmatrix} 0.5 & 0 & 1 & 0 \end{bmatrix}^\top. \quad (35)$$

8 Constructing P

Stacking the stable eigenvectors as columns, in the order $(0.5, 0.2)$, gives

$$\begin{bmatrix} X_{11} \\ X_{12} \end{bmatrix} = \begin{bmatrix} 0.5 & 0 \\ 0 & 0.2 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad X_{12} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad \Delta_1 = \begin{bmatrix} 0.5 & 0 \\ 0 & 0.2 \end{bmatrix}. \quad (36)$$

Since $X_{12} = I$, its inverse is also I , and (7) gives

$$P = X_{12} \Delta_1 X_{12}^{-1} = \Delta_1 = \begin{bmatrix} 0.5 & 0 \\ 0 & 0.2 \end{bmatrix}. \quad (37)$$

Here the eigenvectors turned out aligned with the axes, so X_{12} reduced to the identity and P equals the diagonal matrix of stable eigenvalues. In a general case X_{12} would have to be inverted, but the procedure is identical.

9 Verification

Since P is diagonal,

$$P^2 = \begin{bmatrix} 0.25 & 0 \\ 0 & 0.04 \end{bmatrix}. \quad (38)$$

Then

$$AP^2 = \begin{bmatrix} 3 & 0.5 \\ 7 & 1 \end{bmatrix} \begin{bmatrix} 0.25 & 0 \\ 0 & 0.04 \end{bmatrix} = \begin{bmatrix} 0.75 & 0.02 \\ 1.75 & 0.04 \end{bmatrix}, \quad (39)$$

$$BP = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 0.5 & 0 \\ 0 & 0.2 \end{bmatrix} = \begin{bmatrix} 0.5 & 0 \\ 1 & 0.2 \end{bmatrix}, \quad (40)$$

so that

$$AP^2 + BP + C = \begin{bmatrix} 0.75 & 0.02 \\ 1.75 & 0.04 \end{bmatrix} + \begin{bmatrix} 0.5 & 0 \\ 1 & 0.2 \end{bmatrix} + \begin{bmatrix} -1.25 & -0.02 \\ -2.75 & -0.24 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}. \quad (41)$$

The matrix P solves the equation, as required. Starting only from A , B , C and applying the generalized-eigenvalue method, we recovered

$$P = \begin{bmatrix} 0.5 & 0 \\ 0 & 0.2 \end{bmatrix}. \quad (42)$$

A remark on uniqueness. The quartic (27) has four roots, and any choice of two of them yields a matrix satisfying (1); the equation therefore has several solutions. The stability criterion $|\lambda| < 1$ selects the economically relevant one, and when more than n stable roots exist the convention is to take the n smallest in modulus. Here exactly two roots are stable, so P is pinned down without ambiguity.